Upper Division Tutoring Program Content Review

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Final Review

Inner Product Spaces

Definition 1 (Inner Product)**.** We can give a vector space V an **inner product** $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ for which

(i) $\langle v, v\rangle \geq 0$, (ii) $\langle v, v\rangle = 0 \iff v = 0$, (iii) $\langle u+v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$, (iv) $\langle \lambda u, v\rangle = \lambda \langle u, v\rangle$ ($\lambda \in \mathbb{F}$), (v) $\langle u, v\rangle = \overline{\langle v, u\rangle}$.

Definition 2 (Norm)**.** The **norm** of v is $||v|| = \sqrt{\langle v, v\rangle}$. Notably, $||\lambda v|| = |\lambda| ||v||$ for $\lambda \in \mathbb{F}$.

Definition 3 (Orthogonal)**.** Two vectors $v, w \in V$ are **orthogonal** if $\langle \lambda v, w\rangle = 0$.

Definition 4 (Orthonormal)**.** A list of vectors $e_1, \dots, e_n$ is **orthonormal** in V if $\langle e_i, e_j\rangle = \delta_{ij}$ for all $i, j = 1, \dots, n$.

Remark. An orthonormal list of vectors is linearly independent (you should prove this!)

Remark. Given an orthonormal basis $e_1, \dots, e_n$ of V, then any $v \in V$ can be written $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$!!!

Proposition 1 (Gram-Schmidt)**.** We can construct an orthonormal basis from a random basis $v_1, \dots, v_n$ via the following useful but annoying inductive process: let $f_1 = v_1$, $f_k = v_k - \frac{\langle v_k, f_1}{||f||^2} f_1 - \cdots - \frac{\langle v_k, f_{k-1}\rangle}{||f_{k-1}||^2} f_{k-1}, e_k = \frac{f_k}{||f_k||}$.

Corollary 1. Gram-Schmidt also works on linearly independent lists which aren't bases. Notably, $\text{span}(v_1, \dots, v_k) = \text{span}(e_1, \dots, e_k)$ for any $k = 1, \dots, n$. So if each $(v_1, \dots v_k)$ is T-invariant for some $T \in \mathcal{L}(V)$, then T has an upper-triangular matrix with respect to some orthonormal basis. (Recall this happens if and only if the minimal polynomial of T can be written $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_i \in \mathbb{F}$!)

Riesz Representation

Theorem 1. For a linear functional $\phi \in V^*$, there is a unique vector $v \in V$ so $\phi(u) = \langle u, v\rangle$.

Definition 5 (Orthogonal Complement)**.** For a subspace $U < V$, $U^\perp = \{v \in V : \langle v, u\rangle = 0 \text{ for all } u \in U\}$ is the **orthogonal complement** of U. Crucially, $V = U \oplus U^\perp$ and $U = (U^\perp)^\perp$. (Try proving this!!!)

Definition 6 (Projection)**.** Given $U < V$, any $v = u + w$ for $u \in U, w \in U^\perp$. This induces the **projection** $P_U(v) = u$.

Remark. For a subspace $U < V$, $||v - P_U v|| \leq ||v - u||$ for any $u \in U, v \in V$. Equality holds if and only if $u = P_U v$.

Spectral Theorem

Definition 7 (Adjoint)**.** For $T \in \mathcal{L}(V, W)$, the **adjoint** map $T^* \in \mathcal{L}(W, V)$ is defined by $\langle Tv, w\rangle_W = \langle v, T^* w\rangle_V$.

Remark. Given orthonormal bases (e_i) of V, (f_i) of W, $M(T^*, (f_i), (e_i))$ is the **conjugate transpose** of $M(T, (e_i), (f_i))$.

Remark (Niceness of Adjoint)**.**

$$(S + T)^* = S^* + T^*, \quad (T^*)^* = T, \quad (ST)^* = T^* S^*, \quad (T^*)^{-1} = (T^{-1})^*, \quad \ker T^* = \text{range} T^\perp.$$

Definition 8 (Self-Adjoint, Normal)**.** An operator $T \in \mathcal{L}(V)$ is **self-adjoint** if $T = T^*$ and **normal** if $TT^* = T^* T$.

Theorem 2 (Real Spectral Theorem)**.** If $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$, then

T is self-adjoint $\iff$ T is diagonalizable with an orthonormal basis of V $\iff$ V has an orthonormal basis of T-eigenvectors.

Theorem 3 (Complex Spectral Theorem)**.** If $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$, then

T is normal $\iff$ T is diagonalizable with an orthonormal basis of V $\iff$ V has an orthonormal basis of T-eigenvectors.

Practice Exercises

1. Use Gram-Schmidt to orthogonalize the basis $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

2. Let $\mathcal{P}_3(\mathbb{R})$ be the vector space of real polynomials of degree at most 3 equipped with the inner product
$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)dx.$$
   (a) Verify that the above inner product is indeed an inner product.
   (b) Find an orthonormal basis for $\mathcal{P}_2(\mathbb{R})$.

3. (**Holtz 2023**). If unit vectors $(e_i)_1^n$ satisfy $||v|| = \sum_1^n |\langle v, e_i \rangle|^2$ for all $v \in V$, show e_i is an orthonormal basis.

4. (Axler 6.C.12). Find a polynomial $p \in \mathbb{P}_3(\mathbb{R})$ such that $p(0) = 0$, $p'(0) = 0$, and
$$\int_{-1}^{1} (2 + 3x - p(x))^2 \, dx$$
is as small as possible.

5. (**T/F**): A finite-dimensional vector space is always a direct sum of its eigenspaces.

6. Let V be a finite dimensional complex inner product space and let $T \in \mathcal{L}(V)$ be a normal operator. Let $\Lambda \in \mathbb{R}$ be defined as
$$\Lambda = \max\{|\lambda_1|, \ldots, |\lambda_m|\}$$
where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T. Show that $||Tv|| \leq \Lambda ||v||$ for all $v \in V$.

7. **(Holtz 2023)** Let T be an normal operator on a complex finite-dimensional inner product space V whose eigenvalues are $\lambda_1, \ldots, \lambda_k$. Define
$$R = \max\{|\lambda_1|, \ldots, |\lambda_k|\}.$$
Prove that $||Tv|| \leq R||v||$ for any vector $v \in V$.